

Benjamin Chen

benjaminchen517@gmail.com | (949) 278-6150 | Irvine, CA | LinkedIn: [Benjamin Chen](#)

Civil and Environmental Engineering graduate student specializing in systems analysis, engineering modeling, and data-driven problem-solving. Experience in building management systems, analytical workflows, MATLAB modeling, and technical reporting, with a focus on evaluating system performance and supporting engineering decision-making.

EDUCATION

University of California, Irvine *September 2025 – June 2026*

Master of Engineering in Civil and Environmental Engineering, Sustainable Infrastructure Track

University of Illinois Urbana-Champaign *August 2020 – May 2024*

Bachelor of Science in Agricultural and Biological Engineering

Minor in Sustainability, Energy, and Environment Fellows Program

Relevant Coursework: Data Science Programming & Optimization, Engineering Measurement Systems, Engineering Risk & Uncertainty, Renewable Energy Systems, Engineering for Disaster Resilience

PROFESSIONAL EXPERIENCE

Engineering Intern *Cataño, Puerto Rico*

Caras con Causa *May – August 2023*

- Constructed a ~400 m² coastal restoration nursery supporting wetland ecosystem rehabilitation and climate resilience
- Implemented a climate-responsive irrigation system using a Hunter smart controller and automated scheduling
- Applied research on mangrove nursery design to inform site layout, irrigation strategy, and plant cultivation methods

Disaster Resilience Project *September 2022 – January 2023*

- Conducted literature review on hydroponic nursery design and ebb-and-flow irrigation for mangrove propagation

National Chi Nan University - Office of International Affairs *Puli, Taiwan*

Project Assistant *September 2024 – March 2025*

- Organized 200+ agreements and MOUs, executed associated programs, and secured 10+ new partnerships

PROJECTS

Quantifying Laboratory Space Equipment Loads | Capstone Project *Irvine, CA*

Engineering Team Lead, project sponsored by Altura Associates *December 2025 – June 2026*

- Led a 5-person engineering team developing a scalable methodology for laboratory HVAC load estimation
- Designed analysis workflow using Python and the SkySpark API to parse BMS sensor data and analyze performance
- Scaled the workflow across 250+ HVAC zones and laboratory spaces to evaluate equipment load requirements
- Validated model outputs against equipment nameplate ratings, achieving estimates within 15% of design values

Thresher Safety Optimization | Capstone Project *Urbana-Champaign, IL*

Engineering Consultant, project sponsored by SAYeTECH *February 2024 – May 2024*

- Conducted safety and mechanical risk assessment of ST-6000 Gen II Multi-Crop Thresher, identifying operational hazards including exposed moving components, vibration transmission, and noise exposure
- Analyzed drivetrain rotational inertia and evaluated conventional and dual-mass flywheels for system performance
- Contributed to CAD design of bucket elevator and operator platform to reduce direct exposure to vibrating machinery
- Evaluated cost and market feasibility analysis to assess competitiveness of proposed design modifications

Sol Systems Recycler Outreach | Capstone Project *Urbana-Champaign, IL*

Sustainability Consultant, project sponsored by Sol Systems *September 2023 – December 2023*

- Interviewed 14 photovoltaic recyclers and off-takers across 4 states to evaluate recycling and supply chain practices
- Created 4 portfolios detailing recommendations, recycler information, and notable recycling legislation and regulations

Anteater Tetris Club at UCI *Irvine, CA*

Founder, Graduate Advisor *October 2025 – Present*

- Trained and mentored club officers on event operations and logistics to support organizational planning
- Supported growth to 100+ members and consistent 12+ meeting attendance with active online engagement
- Coordinated captains of 3 competitive Tetris teams (15 starters, 7 subs) in the 54-team collegiate league

TECHNICAL SUMMARY

Programming: Python, MATLAB, R, JavaScript

Modeling, Simulation & Analytics: Simulink, SkySpark, Project Haystack, eQUEST, ArcGIS

Engineering Systems & Software: HVAC, BMS Systems, SimaPro, Sphera (GaBi), Adobe Creative Suite